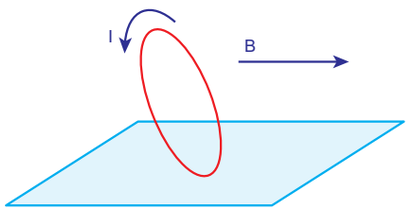


Solution



When the current is passed through the loop, the torque is produced. If the torque acting on the loop is increased then the loop will start to rotate. The loop will start to lift if and only if the magnitude of magnetic torque due to current applied equals to the gravitational torque as shown in Figure

$$\tau_{magnetic} = \tau_{gravitational}$$

$$IAB = mgR$$

But $p_m = IA = I(\pi R^2)$

$$\pi I R^2 B = mgR$$

$$\Rightarrow I = \frac{mg}{\pi RB}$$

The current estimated using this equation should be applied so that one edge of loop is lifted of the surface.

A4.1 MOTIONAL EMF FROM FARADAY'S LAW

Let us consider a rectangular conducting loop of width l in a uniform magnetic field $\vec{B}$ which is perpendicular to the plane of the loop

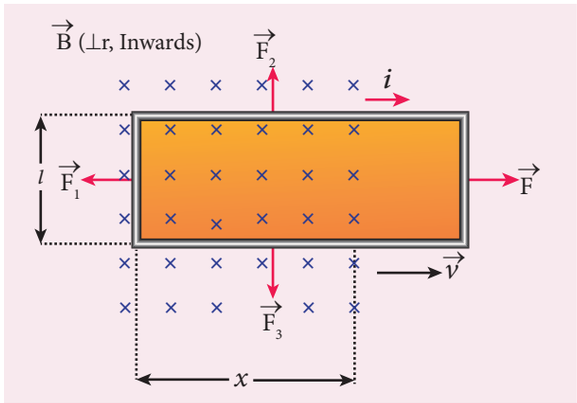


Figure A4.1 Motional emf from Faraday's law

and is directed inwards. A part of the loop is in the magnetic field while the remaining part is outside the field as shown in Figure A4.1.

When the loop is pulled with a constant velocity $\vec{v}$ to the right, the area of the portion of the loop within the magnetic field will decrease. Thus, the flux linked with the loop will also decrease. According to Faraday's law, an electric current is induced in the loop which flows in a direction so as to oppose the pull of the loop.

Let x be the length of the loop which is still within the magnetic field, then its area is lx. The magnetic flux linked with the loop is

$$\Phi_B = \int_A \vec{B} \cdot d\vec{A} = BA\cos\theta = BA$$

Here $\theta = 0^\circ$ and $\cos 0^\circ = 1$

$$\Phi_B = Blx \qquad \text{(A4.1)}$$

As this magnetic flux decreases due to the movement of the loop, the magnitude of the induced emf is given by

$$\varepsilon = \frac{d\Phi_B}{dt} = \frac{d}{dt}(Blx)$$

Here, both B and l are constants. Therefore,

$$\varepsilon = Bl\frac{dx}{dt} = Blv \qquad \text{(A4.2)}$$

Appendix-1.indd 325 14-02-2020 20:14:43